CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 24: Solution of Right Triangles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. In triangle ABC, $\angle B = 90^\circ$, $\angle A = 30^\circ$ and $AC = 14$ cm. Find BC and AB.

2. In triangle PQR, $\angle Q = 90^\circ$, $\angle R = 45^\circ$ and $QR = 9$ cm. Find PQ and PR.

3. In triangle ABC, $\angle B = 90^\circ$, $AB = 6\sqrt{3}$ cm and $BC = 6$ cm. Find $\angle A$ and AC.

4. A ladder 8 m long makes 30° with the ground. How high up the wall does it reach?

5. Find the altitude of an equilateral triangle of side 12 cm.

6. In triangle ABC, AD is perpendicular to BC, $\angle B = 60^\circ$, $\angle C = 45^\circ$ and $AD = 9$ cm. Find BC.

7. A rhombus has side 8 cm and one angle 120°. Find its diagonals.

8. A rectangle has diagonal 20 cm, which makes 30° with the longer side. Find the sides.

9. A chord of a circle of radius 6 cm subtends 60° at the centre. Find its length.

10. The top of a tower is seen at 45° from a point 30 m from its foot. Find the height of the tower.

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Answer Key

1. 7 cm and $7\sqrt{3}$ cm
2. 9 cm and $9\sqrt{2}$ cm
3. $30°$; 12 cm
4. 4 m
5. $6\sqrt{3}$ cm
6. $3\sqrt{3} + 9$ cm, about 14.2 cm
7. 8 cm and $8\sqrt{3}$ cm
8. $10\sqrt{3}$ cm and 10 cm
9. 6 cm
10. 30 m

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